

ReQTable: Square tabletop display that provides dual-sided mid-air images to each of four users

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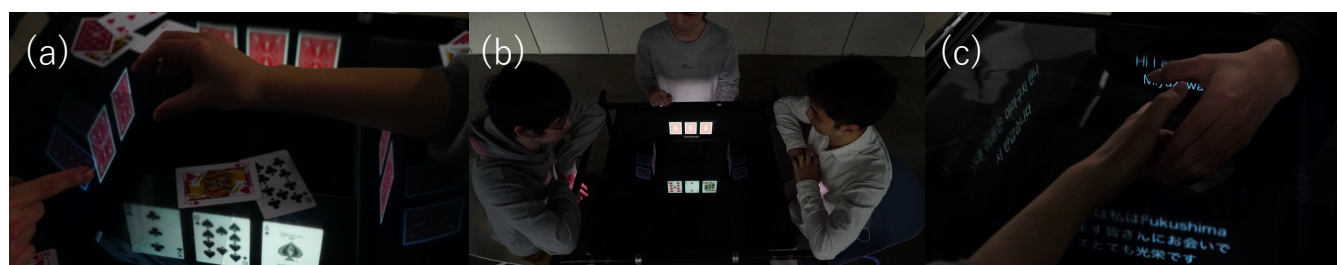


Figure 1: Applications using the ReQTable. (a) and (b) Playing card games with multiple players using physical and virtual cards. (c) Communication with foreign people around a table: the subtitles help the participants talking and they can shake hands without hitting the display.

ABSTRACT

We propose “ReQTable”, an optical system displaying dual-sided mid-air images to each of four users with less stray light. Dual Slit Mirror Arrays (dual SMAs) can produce a mid-air image that multiple people can view simultaneously without wearing special equipment. Dual SMAs are rectangular and can theoretically display mid-air images in the four surrounding directions. However, they also produce unwanted light, called stray light, which is especially noticeable when mid-air images are displayed in some directions. It is difficult for users to identify stray lights, and they cannot concentrate on viewing only the mid-air images. To improve a user experience, it is necessary to suppress stray light. ReQTable displays mid-air images in four directions with stray light reduction.

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1 INTRODUCTION

A mid-air image has no display frame, and enhances the sense of mixed reality by being displayed with objects used in daily life. Moreover, mid-air images on a table create an environment where users can interact with characters in their real lives and play games with others [Chan et al. 2008; Yoshida 2015]. Dual SMAs, made of two orthogonal SMAs, are optical devices that can produce mid-air images [International Electrotechnical Commission 2020]. Dual SMAs can form undistorted mid-air images by light reflections at a position that is symmetrical to a light source with respect to it. Dual SMAs can theoretically display mid-air images in four directions around the device, because of their rectangular shape. However, they have the disadvantage of generating stray light, which is particularly noticeable when displaying mid-air images in multiple directions. Stray light confuses mid-air images, and interferes with the user’s appreciation of them. Therefore, it is especially necessary to suppress stray light when generating mid-air images in some directions using dual SMAs.

Dual SMAs are two-layered. Therefore, the images produced by them are classified into four types based on the number of reflections (Table 1) [Okawa et al. 2016]. The light reflected an odd number of times in each layer constitutes the mid-air image and rest is stray light. There are two main methods for suppressing undesired light in mid-air imaging optics. The first method involves the use of polarizers. Light with an even or odd reflection frequency can be selectively suppressed by passing it through a polarizer before and after it is reflected. Osada and Bao [2021] suppressed ghost images by using circular polarizers. The second method is

Table 1: Classification of images produced through dual SMAs by a number of reflections

		SMA (1st layer)	
		odd times	even times
SMA (2nd layer)	odd times	Mid-air images	Ghost images
	even times	Ghost images	Transmitted light

to place view control films (VCFs). A VCF can block light only in certain directions. Katsumoto et al. [2016] reduced the transmitted light in two opposite directions by using VCFs. However, there has been no study on displaying mid-air images for each of the four users while suppressing all types of stray light. In this paper, we propose an optical system called ReQTable to enhance mid-air images from four positions by reducing stray light by using linear polarizers and VCFs.

2 IMPLEMENTATION

The proposed system is designed to display dual-sided mid-air images for each of the four users by using dual-sided displays (Fig. 2 (a)). In this system, linear polarizers and VCFs are used for the suppression of ghost images and transmitted light respectively as shown in Fig. 2 (b). Dual-sided displays as light sources are placed under the dual SMAs. In this case, the light rays from the back of the dual-sided displays forms mid-air images for each of the users.

The light rays from the display are linearly polarized by the first polarizer. Next, the rays that form the ghost images are reflected an odd number of times in the dual SMAs. Therefore, the direction of their polarization changes, and they cannot pass through the second polarizer. In contrast, transmitted light rays and light rays forming mid-air images are reflected an even number of times in the dual SMAs. Their direction of polarization is different from that of the rays that become ghost images. Therefore, they can pass through the second polarizer. Polarizers cannot suppress the transmitted light, but unlike mid-air images, it is the light that emerges from the front of the dual-sided displays. By modifying the VCF placement method of Katsumoto et al., we orthogonally placed VCFs on the second polarizer. Only the transmitted light in the two opposite directions along the two orthogonal axes cannot pass through the VCFs. By combining the two methods of using polarizers and the VCFs, we can selectively suppress other lights, retaining only those that produce a mid-air image.

This system is used for the table surface. When the mid-air images of the cards are presented in four positions, it is obvious that the stray light in the proposed system is less than as compared to that in the dual SMAs (Fig. 3 (a) and (b)).

3 APPLICATIONS

ReQTable can display different mid-air images on the front and back of the four sides. Users of the system can work or play using the mid-air images. ReQTable helps us perform collaborative work on a table (Fig. 1 (b)). As it is only the mid-air images that have no display frames, the users can communicate physically, such as shake hands, and can easily place objects on the table surface. A four player game can be played using ReQTable (Fig. 1 (a)). The

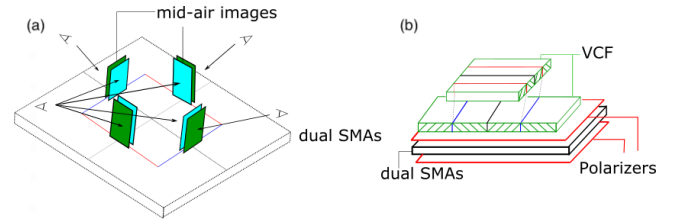


Figure 2: Overview of proposed method. (a) Positions of mid-air images and users. (b) Polarizers sandwiching dual SMAs. VCFs are placed on the top of the sandwich structure

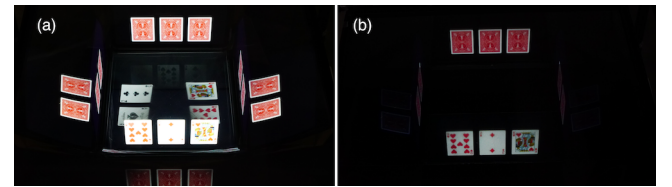


Figure 3: The effect of the proposed method. (a) Original mid-air images. (b) Mid-air images in ReQTable.

mid-air images are dual-sided. Each player can only see their hand of cards while the other players' cards are hidden, i.e., they only see the backs of the other three players' cards. Users will see the content of others' cards through stray light, but ReQTable can prevent them from doing so. Therefore, players can play various card games such as poker, without revealing their cards to others.

4 DEMONSTRATION

As mentioned earlier, participants can play card game using ReQTable. Players can touch the mid-air images of the displayed cards to perform operations. We believe that users can experience mixed reality through this demonstration.

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